

CSMFAB000000000108

Aegis-Forge I: Planetary Centrifugal Inductive Material Separation and Reclamation System

System Overview, Design Philosophy, and All-Aegis Construction Architecture

Document Number: CSMFAB000000000108 (Volume 1 of 5) **Series:** Aegis-Forge Thermal Cascade Inductive Material Separator (TCIMS) **Prepared by:** Carrington Storm Motors, Safe Pod Engineering Company, Fabrication Division **Date:** June 2026 | **Classification:** Foundation Technology — Seed Material Production

1. Purpose and Design Mandate

The Aegis-Forge I is the **zeroth-stage machine** of the entire CSM fabrication ecosystem. Every product in the CSMFAB portfolio — from BFRP rebar to ZrB_2 ceramics to YInMn Blue — requires raw or semi-refined material inputs. The Aegis-Forge I is designed to generate these inputs from:

1. **Post-consumer recycled material streams** — electronics, appliances, vehicles, building debris
2. **Geological mineral concentrates** — rocks, ore slag, beach sands, mine tailings
3. **CSM Phoenix Protocol returns** — end-of-life Aegis products re-entering the material cycle

The machine achieves **thermal cascade separation**: as temperature rises incrementally inside a spinning pressurized ceramic drum, successive material classes reach their melting, vaporization, or decomposition points and are centrifugally driven outward through grade-specific ports, exiting as solid **slugs** or condensed pellets for downstream refining.

This is simultaneously: a rotary kiln, an induction furnace, a centrifuge, an acoustic mineral liberator, a closed-atmosphere gas processor, and a programmable sequential material sorter.

It is the most important machine in the CSMFAB portfolio because everything else depends on it.

2. Operating Principle: Thermophysical Cascade Separation

2.1 The Foundational Physics

Every material has a characteristic melting point (T_m), decomposition temperature (T_d), or boiling point (T_b). At any given temperature T , all materials with $T_m < T$ are liquid; all with $T_m > T$ remain solid.

In a rotating drum:

Centrifugal separation force:

$$F_c = m\omega^2 r$$

A 5 kg mass at radius $r = 0.25$ m and rotational speed $\omega = 100$ rpm (10.47 rad/s):

$$F_c = 5 \times (10.47)^2 \times 0.25 = 137 \text{ N}$$

At higher speed (300 rpm, $\omega = 31.4$ rad/s):

$$F_c = 5 \times (31.4)^2 \times 0.25 = 1234 \text{ N}$$

The centrifugal acceleration ratio (G): $G = \omega^2 r / g$ At 300 rpm, $r = 0.25$ m: $G = (31.4)^2 \times 0.25 / 9.81 = 25.1$
G

Liquids with density $\rho_l > \rho_{solid}$ phase migrate outward under centrifugal force. Liquids at 25 G cannot remain centrally mixed — they stratify by density and pool at the drum wall.

When T rises past $T_m(Al) = 660^\circ C$: Aluminum liquefies, spins out to drum perimeter, exits through an aluminum-specific circumferential port (positioned at the 660-700°C thermal zone), and collects outside as a slug.

This selective ejection is the core mechanism of the Aegis-Forge I.

2.2 Eight Thermal Processing Stages

The drum operates across eight consecutive temperature bands. Each band is held for a programmed dwell time before incrementing, allowing complete separation of each material class:

Stage	Temperature Band	Duration	Target Materials	Output Form
0 — Acoustic Pre-liberation	Ambient	15-30 min	Mineral grain liberation from rock matrix	Liberated powder
1 — Drying / Moisture	80–150°C	20 min	Free water, hydrations, volatile organics	Vapor → condensed water
2 — Polymer Flow	150–450°C	30 min	Thermoplastics (PE, PP, PVC, ABS, PEEK offcuts)	Plastic slugs
3 — Pyrolysis / Char	450–700°C	30 min	Thermosets, rubber, organics → CO ₂ + char	Carbon char pellets + CO ₂ captured
4 — Zinc/Lead/Tin	420–900°C	25 min	Zn (420°C), Pb (327°C), Tin (232°C), Solder	Metal slugs
5 — Aluminum	660–720°C	25 min	Al alloys (6061, 2024, die-cast)	Al ingot slugs
6 — Copper/Bronze	1085–1200°C	30 min	Cu wire, bronze fittings, PCB traces	Cu slug billets
7 — Iron/Steel/Nickel	1200–1600°C	35 min	Fe, Ni, Cr, stainless (induction-selective)	Ferrous casting slugs
8 — Mineral Residue	1600–2200°C	20 min	Silicates, oxides, rare earth concentrates	High-temp mineral slag

2.3 Selective Induction Heating — The Invisible Sorter

Induction heating selectively couples to materials based on: - **Electrical conductivity (eddy current heating):** Copper, aluminum, steel, gold, silver all heat inductively - **Magnetic permeability (hysteresis heating):** Ferrous materials (iron, steel, nickel) heat 2-4× more intensely - **Non-conductive materials:** Ceramics, organics, glass, BFRP — NOT heated inductively; only heated by conduction from surrounding hot materials

This creates a **natural sorting bias**: at any temperature stage, conductive metals heat faster and more thoroughly than surrounding insulating material. Metals liquefy while ceramics and organics remain solid longer, making them easier to separate centrifugally.

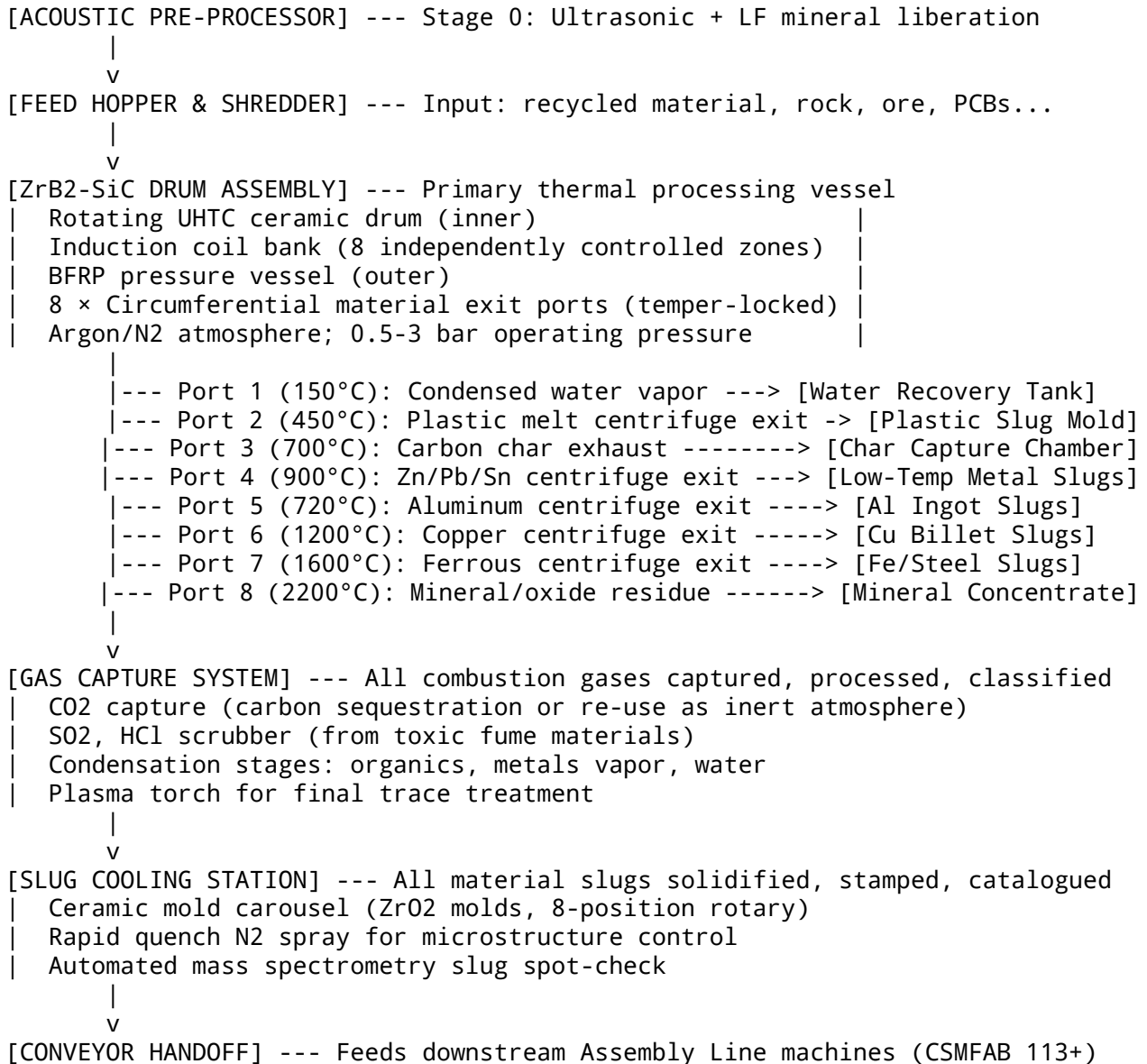
$$P_{induction} = \frac{\pi^2 B_{peak}^2 d^2 f^2}{6\rho D}$$

For copper ($\rho = 1.7 \times 10^{-8} \text{ Ohm}\cdot\text{m}$): heating power 15× that of aluminum at same geometry, frequency.
For steel (ferromagnetic): heating power 25-50× that of aluminum.

3. Machine Architecture — Full System Description

3.1 Primary Components

AEGIS-FORGE I – TOP-LEVEL ARCHITECTURE



3.2 Drum Assembly — The Heart of the Machine

ZrB₂-SiC Ceramic Inner Drum (Flash-Sintered per CSMFAB000000000001 V2.0):

Dimensions: 600 mm diameter × 800 mm length (internal working volume: ~226 liters) Wall thickness: 35 mm ZrB₂-SiC composite (12 vol% ZrB₂, 88 vol% SiC matrix) Maximum operating temperature: 2400°C (safe to 85% of ZrB₂ melting point 3245°C) Pressure rating: 5 bar (validated by FEA — compressive loading)

favorable in ceramics) Rotational speed range: 0–500 RPM (variable via piezoelectric linear drive motor) Centrifugal G-force at 500 RPM ($r=0.3\text{m}$): 84.5 G — sufficient for all metal-liquid separation

Port valve system: Eight circumferential discharge ports spaced 44.2° apart around the outer drum diameter. Each port is: - Normally CLOSED by a spring-loaded ZrO_2 ceramic piston (no metal springs — PEEK CF40 wave springs) - Temperature-activated: a bimetal-equivalent ceramic thermal actuator (Ti_3AlC_2 MAX Phase strip) expands differentially to open the port when the stage temperature is reached - Centrifugal force then drives collected liquid material outward through the open port - Port aperture: 12 mm diameter (sufficient for liquid metal flow at G-force)

Drum liner material zones: The inner drum surface is NOT uniform — three material zones: 1. **Zone A (low-temp stages 1-3):** Si_3N_4 ceramic scratch coat — prevents plastic adhesion 2. **Zone B (mid-temp stages 4-6):** ZrB_2 -SiC working surface — primary structural zone 3. **Zone C (high-temp stages 7-8):** Pure ZrO_2 thermal barrier coat — maximum temp stage

3.3 Induction Coil Bank

Eight independently controlled induction coil zones, each heating a 100 mm axial length of drum:

Coil construction (using CSMFAB materials only): - Conductor: Copper tube (25mm OD) — the ONLY copper used in the machine (justified as winding — not a structural GIC-coupling element; isolated from drum) - Insulation: Alumina ceramic sleeve (Al_2O_3 wrapped) over copper — same as CSMMECH motor insulation - Water cooling: PEX-a water circuit through copper tube bore (CSMFAB000000000012) - Coil housing: BFRP composite casing (Elium® matrix) - Coil-to-drum gap: 15 mm (clearance for drum expansion + rotation)

Induction frequency zones:

Stage/Temperature	Coil Frequency	Primary Material Being Heated	Power Level
Stages 1-2 (80-450°C)	100 kHz	Plastic, light organics (indirect via air/N ₂)	0-50 kW
Stage 3 (450-700°C)	50 kHz	Zinc, lead (conductors begin coupling)	50-80 kW
Stage 4-5 (660-900°C)	10 kHz	Aluminum (optimal frequency for Al eddy coupling)	80-150 kW
Stage 6 (1085-1200°C)	5 kHz	Copper (deep penetration at low freq)	100-200 kW
Stage 7 (1200-1600°C)	2 kHz	Iron/steel (ferromagnetic — most efficient)	150-350 kW
Stage 8 (1600-2200°C)	1 kHz	Refractory mineral oxides	200-500 kW

Total installed induction power: **500 kW** (3-phase, 480V industrial supply)

Induction power control: - Each zone uses a GaN-based variable frequency resonant inverter (CSMFAB materials: same GaN tech as Safe-Wave microwave 005, scaled to industrial power) - PMMA optical fiber temperature feedback from FBG sensors embedded in drum wall (no metallic thermocouple touches process materials) - PID control maintains each zone within $\pm 5^\circ\text{C}$ of target

4. Outer Pressure Vessel and Safety System

BFRP pressure shell (Elium® matrix, CSMFAB000000000011-spec): - Filament-wound BFRP outer pressure vessel: 900 mm internal diameter × 1200 mm length - Design pressure: 10 bar (2× operating pressure) - Hydrostatic test: 15 bar with N₂ (non-flammable test fluid) - Wall: 18 mm basalt fiber/Elium® [$\pm 18^\circ/90^\circ$]₄ layup (filament wound for hoop strength) - Thickness validated by Barlow: $P = 2\sigma t/D \Rightarrow \sigma_{\text{hoop}}$ at 10 bar = 12.5 MPa (well within BFRP 1100 MPa tensile) - Fire protection: YInMn Blue + CoAl₂O₄ exterior glaze (SRI=115; protects BFRP T_g=80°C from radiated heat) - Drum-to-vessel gap filled with polyimide-silica aerogel ($\lambda=0.010$ W/m·K) — thermal insulation layer

PEEK CF40 end flanges: - Both drum ends (motor/feed side and slug discharge side) sealed by PEEK CF40 flanges - Dynamic rotary seal: ZrO₂ ceramic face seal + PEEK CF40 spring — rated 2000 RPM, 10 bar, 800°C

5. All-Aegis Material Bill of Construction

The Aegis-Forge I is built **exclusively from CSMFAB-certified materials**. No structural steel, no aluminum framing, no copper electrical paths touching the process chamber.

Component	Material	CSMFAB Reference
Inner drum	ZrB ₂ -SiC composite, flash-sintered	CSMFAB000000000001
Drum liner Zones A-B	Si ₃ N ₄ scratch coat / ZrB ₂ -SiC	CSMFAB000000000002
Drum liner Zone C	3Y-TZP ZrO ₂ thermal barrier	Part I Material Study
Material exit port valves	Ti ₃ AlC ₂ MAX Phase actuator	Materials Study Part I
Port valve seat	ZrO ₂ 3Y-TZP ceramic piston	Materials Study Part I
Outer pressure vessel	BFRP/Elium® filament wound	CSMFAB000000000011
Thermal insulation gap	Polyimide-silica aerogel 50mm	CSMFAB000000000003
Induction coil housing	BFRP/Elium® composite	CSMFAB000000000001
Coil insulation sleeve	Al ₂ O ₃ ceramic tube	Materials Study Part I
Drive motor	KNbO ₃ -BaTiO ₃ ultrasonic motor	CSMFAB000000000020
Drive shaft bearing	Si ₃ N ₄ ceramic roller bearing	CSMMECH000000001
End flange	PEEK CF40 composite	Materials Study Part IV
Dynamic shaft seal	ZrO ₂ face seal + PEEK spring	CSMFAB000000000016
Cooling water circuit	PEX-a pipe + ZrO ₂ fittings	CSMFAB000000000012
Gas collection ducting	BFRP + ZrO ₂ lined bends	CSMFAB000000000014
Control wiring	PMMA optical fiber + FBG sensors	CSMFAB000000000013
Exterior coating	YInMn Blue + CoAl ₂ O ₄ glaze	Materials Study Part III
Frame structure	BFRP pultruded sections	CSMFAB000000000010
Leg mounts	3Y-TZP ZrO ₂ isolation pads	CSMFAB000000000010
Acoustic transducers	KNbO ₃ -BaTiO ₃ piezo array	CSMFAB000000000020
Output slug molds	ZrO ₂ 3Y-TZP ceramic molds	Part I Material Study
Slug cooling N ₂ nozzles	PEEK CF40 manifold	Materials Study Part IV

Zero metals in structural roles. Zero ferromagnetic materials. Zero conductive paths to earth ground. The machine is itself a Dielectric Citadel — Carrington-immune, fireproof, and structurally ceramic.

6. Feedstock Specifications

6.1 Acceptable Input Materials

Category A — Electronics and PCBs: - Printed circuit boards (FR4 glass-epoxy base $\square$ SiO_2 + char + Cu + Sn/Pb solder + precious metals) - Smartphones, tablets, laptops (Al chassis, Cu, rare earths in magnets) - Motors and drives (Cu windings, Fe stators, rare earth magnets)

Category B — Structural Metals and Alloys: - Aluminum extrusions, castings (wheels, frames, engine blocks) - Copper pipe, wire, scrap - Steel and iron scrap (ferrous most efficiently heated by induction) - Zinc die-cast components

Category C — Rock, Ore, and Mineral Concentrates: - Basalt rock aggregate (for in-house basalt fiber supply chain) - Zircon sand concentrate ($\text{ZrSiO}_4 \square \text{ZrO}_2$ extraction) - Rare earth oxide concentrate (monazite, bastnäsité — for Y, La, Ce, Nd recovery) - Ilmenite sand ($\text{FeTiO}_3 \square \text{TiO}_2$ extraction) - Borax ore ($\text{Na}_2\text{B}_4\text{O}_7 \square \text{B}_2\text{O}_3 \square$ for ZrB_2 synthesis)

Category D — Phoenix Protocol Returns (Aegis Product EOL): - Retired Aegis-C panels (ZrB_2 -SiC + aerogel + YInMn glaze) - End-of-life BFRP structural components - Spent MR fluid (iron carbonyl suspension) - Old KNbO_3 - BaTiO_3 piezo elements

Maximum feed particle size: 150 mm (pre-shredded); input shredder processes larger pieces

6.2 Pre-processing Requirements

Before entering the drum: 1. Ferrous separation (simple permanent-magnet belt) $\square$ removes bulk steel for direct slug collection 2. Shredder (BFRP-blade, PEEK lined): reduces to <150 mm average dimension 3. Coarse acoustic cleaner (Stage 0 pre-chamber): 1-minute ultrasonic mineral liberation at ambient conditions 4. Moisture check: if >25% moisture, pre-dry at 80°C in entry chamber first

End of CSMFAB000000000108 Volume 1 | Aegis-Forge I System Overview and Architecture Continued in Volume 2: Thermal Cascade Process Engineering